

Wenjun LUO

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EDUCATION

South China University of Technology, B.Sc. in Information And Computing Science Sept 2021 – July 2025

- **GPA:** 87.82/100 (3.7/4.0).
- **Major Courses:** Mathematics Analysis, Advanced Algebra, Programming in C++, Ordinary Differential Equation, Data Structure, Probability Theory, Matrix Computations, Computational Intelligence, etc.

University of Macau, Ph.D. in Electromechanical Engineering Aug 2025 – Present

- **Research Interest:** Safe-critical control and autonomous decision-making of intelligent systems.
- **Supervisor:** Under the supervision of Prof. Jason Jinrong LIU.

Publications

- P. Cheng, **W. Luo**, J. Li, C. Liu, J. J. R. Liu, and Z. Feng, "Fixed-time disturbance observer-based prescribed performance control of rigid spacecraft," IEEE Transactions on Aerospace and Electronic Systems, 2025. (**JCR Q1**)
- P. Cheng, **W. Luo**, C. Liu, J. Li, J. J. R. Liu, D. Li, and Y.-J. Liu, "Event-triggered saturation-tolerant prescribed control of rigid spacecraft with actuator faults," IEEE Transactions on Circuits and Systems I: Regular Papers, 2025. (**JCR Q1**)
- P. Cheng, **W. Luo**, J. J. R. Liu, and Z. Feng, "Dynamic event-triggered fault-tolerant control of rigid spacecraft with prescribed performance," ISA Transactions, 2025. (**JCR Q1**)
- J. Li, **W. Luo**, J. J. R. Liu, and J. Lam, "Twin-based cyber-security control for swarm with aggregation-dispersion ability," in 2025 37th Chinese Control and Decision Conference (CCDC), 2025. (**EI**)
- W. Zhou, S. Chen, J. Li, C. Liu, **W. Luo**, and J. J. R. Liu, "Obstacle avoidance control for cluster-based unmanned aerial vehicle formation with multiple constraints," Franklin Open, 2025. (**JCR Q3**)
- P. Cheng, **W. Luo**, J. J. R. Liu, and Z. Feng, "Fixed-Time Fault-Tolerant Control of Saturated Spacecraft With Full-State Guaranteed Performance", IEEE Transactions on Systems, Man, and Cybernetics: Systems, 2026. (**JCR Q1**)
- J. Li, **W. Luo**, C. Liu, J. J. R. Liu, and J. Lam, "Resilient Consensus for Networked Systems with Hidden Nodes and Edges: Detection, Isolation, and Control", IEEE Transactions on Signal and Information Processing over Networks, 2026. (**JCR Q1**)
- **W. Luo**, Z. Zhang, J. J. R. Liu, and J. Lam, "Learning Discrete-Time Control Barrier Functions via RRT-Guided Sampling and Graph Neural Network", in 2026 45th Chinese Control Conference (CCC), 2026. (**Finalist for the Guan Zhao-zhi Award**)
- W. Zhou, J. Li, **W. Luo**, Z. Zhang, J. J. R. Liu, and J. Lam, "An improved APF Method for UAV Formation Control in Sudden Obstacle Environments", in 2026 15th Asian Control Conference (ASCC), 2026. (**EI**)
- P. Cheng, **W. Luo**, J. J. R. Liu, and Z. Feng, "Predefined-Time Fault-Tolerant Constrained Control of Rigid Spacecraft Without Unwinding", Under Review, 2026.
- J. Li, **W. Luo**, J. J. R. Liu, J. Lam, et al, "A Cloud-Based Security Control for UAV Swarm Against Complex FDI Attacks: From Twin to Physical", Under Review, 2026.
- **W. Luo**, J. Li, Z. Zhang, J. J. R. Liu, et al, "Event-Triggered Disturbance Rejection Taylor-Lagrange Control for Safety-Critical Systems", Under Review, 2026.

RESEARCH EXPERIENCE

Application of difference formats to solving the re-scaled equations and the one-dimensional convection-diffusion equation Jan 2024 - Aug 2024

- Solving the re-scaled equations and first-order convection-diffusion equations arising from practical applications

using the five-point difference format and the backward difference format.

- Modify existing difference formats and promote them to solve more general problems.

Safe-critical control of unmanned aerial vehicles via learning based control barrier function

Aug 2025 - Jan 2026

- An RRT-PID sampling algorithm is proposed for collecting expert demonstrations.
- Using graph neural networks to learn control barrier functions from expert demonstrations, the generalization ability of the algorithm is enhanced.
- The effectiveness and generalization of the proposed algorithm are verified using an LQR-RRT*-CBF controller.

WORK EXPERIENCE

Front-end engineer, Guangzhou Research Tea & Travel Culture Co.

Oct 2023 – Dec 2023

- Assist the technical department in building websites, designing UI, and writing back-end code for software.
- Perform routine maintenance of the company's software.

HONOURS And AWARDS

- **Scholarships (2022):** South China University of Technology Scholarship.
- **Reader's Cup Young People's Literature Competition(2022):** First prize.
- **Mathorcup University Mathematical Modeling Challenge(2023):** First prize at national level.
- **American Collegiate Mathematical Modeling Contest(2023):** S prize.
- **China Undergraduate Mathematical Contest in Modeling (2023):** Second Prize at Provincial Level.
- **The Fourth National Student Psychological Knowledge Contest(2024):** First prize at national level.
- **Merit Student in South China University of Technology(2024).**
- **National Motivational Scholarship (2024).**
- **Young Doctoral Students Talent Support Program of the China Association for Science and Technology (2025).**
- **Finalist for the Guan Zhao-Zhi Award in the 45th Chinese Control Conference (CCC2026).**

PROFESSIONAL SKILLS

- **Languages:** CET 4, CET 6.
- **Software / Programming:** Python, C++, MATLAB, Ubuntu, ROS2, ArduPilot, PX4, VS Code, Git.
- **Algorithms:** Model Predictive Control (MPC), Linear Quadratic Regulator (LQR), A* Path Planning, RRT* Path Planning, Kalman Filter, LSTM, Graph Neural Networks (GNN).